

ATMOSPHERE, AIR & IMPORTANT GASES

Key Terms, Properties, Processes and Environmental Concepts

1. Atmosphere & Layers of Atmosphere

- **Atmosphere:** The protective envelope and cloud of gases surrounding planet Earth.
- **Outer Space:** The region above the atmosphere that is completely free of air.
- **Troposphere:** The innermost layer extending up to 20 km altitude, holding most of the atmospheric mass and where weather phenomena occur.
- **Lapse Rate:** The rate at which air temperature decreases with altitude in the troposphere (6°C per km).
- **Stratosphere:** The atmospheric layer (20–50 km altitude) where aeroplanes fly.
- **Inversion Layer:** The stratospheric region above 25 km where temperature increases with altitude due to UV absorption.
- **Ozonosphere (Ozone Layer):** The atmospheric region (25–40 km altitude) that absorbs harmful solar ultraviolet radiation.
- **Mesosphere:** The middle layer (50–85/90 km altitude) where meteorites burn up.
- **Ionosphere / Thermosphere:** The high-temperature upper layer consisting of ions and free electrons that reflects radio waves.
- **Auroras:** Luminous bands of light in high latitudes caused by energetic solar electrons colliding with N₂ and O₂ molecules.
- **Exosphere:** The outermost atmospheric layer (extending up to 10,000 km) that absorbs harmful ionizing radiation.
- **Atmospheric Pressure:** The downward force per unit area exerted by the weight of air molecules, measured at sea level with a barometer.

2. General Components of Air

- **Active Component (Oxygen):** The component of air (~21% by volume) responsible for respiration and combustion.
- **Inactive Component (Nitrogen):** The inert component (~78% by volume) that dilutes oxygen's reactivity.
- **Fog:** Suspended moisture formed when water vapour condenses onto dust particles during cold temperatures.

3. Oxygen (O₂)

- **Fire Air / Vital Life:** Historical name given to oxygen by Carl Scheele upon preparing it from nitre and mercuric oxide.
- **Catalyst:** A substance (such as manganese dioxide, MnO₂) that alters the rate of a chemical reaction without undergoing permanent chemical change.
- **Downward Displacement of Water:** The collection method used for oxygen gas in the laboratory due to its low solubility.
- **Vapour Density:** The ratio of the mass of a specific volume of a gas to the mass of an equal volume of hydrogen gas under identical conditions.
- **Basic & Acidic Oxides:** Metallic oxides formed with oxygen are generally basic, whereas non-metallic oxides are acidic.
- **Rusting & Rust:** The corrosion of iron in humid air to form brownish, flaky hydrated iron(III) oxide (Fe₂O₃·xH₂O).

- **Rust Prevention Methods:**
 - **Enamelling:** Coating iron surfaces with a mixture of silicates at high temperatures.
 - **Galvanisation:** Coating iron sheets with molten zinc.
 - **Tinning:** Coating iron sheets with a protective layer of tin.
 - **Alloying:** Mixing molten metals to form corrosion-resistant mixtures like stainless steel.
- **Carbogen:** A medical gas mixture of 95% oxygen and 5% carbon dioxide used for patients with breathing difficulties.

4. Carbon Dioxide (CO₂)

- **Fixed Air / Carbonium:** Historical names given to carbon dioxide by Joseph Black and van Helmont.
- **Upward Displacement of Air:** The collection method used for carbon dioxide in the laboratory because it is heavier than air.
- **Dry Ice:** Solid carbon dioxide formed when compressed CO₂ gas expands rapidly through a small nozzle.
- **Lime Water Test:** The diagnostic test where CO₂ turns calcium hydroxide solution milky due to insoluble calcium carbonate precipitation.
- **Fire Extinguisher:** A device utilizing non-combustible, heavy CO₂ gas to smother fires.

5. Nitrogen (N₂)

- **Nitrogen Fixation:** The natural process of converting inert atmospheric nitrogen into soil nitrates via atmospheric lightning or symbiotic bacteria in leguminous root nodules.

6. Hydrogen (H₂)

- **Water-Former:** The literal meaning of hydrogen, named by Antoine Lavoisier.
- **Water Gas:** A gaseous fuel mixture of carbon monoxide (CO) and hydrogen (H₂) generated by passing superheated steam over red-hot coke.
- **Bosch Process:** The industrial manufacturing method used to isolate pure hydrogen gas from water gas.
- **Adsorption / Occlusion:** The accumulation and trapping of hydrogen gas molecules on the surface of metals such as palladium, platinum, or nickel.
- **Hydrogenation:** The catalytic addition of hydrogen gas to unsaturated organic compounds, such as turning liquid vegetable oils into solid fats using a nickel catalyst.
- **Nascent Hydrogen ([H]):** Newly liberated atomic hydrogen possessing powerful reducing capabilities.

7. Air Pollution & Environmental Concepts

- **Pollutants:** Contaminating substances introduced into the air by human activities or natural sources (e.g., CO, SO₂, NO₂, CFCs, Fly Ash, and Suspended Particulate Matter).
- **Acid Rain:** Rainfall contaminated by dissolved sulfur dioxide and nitrogen dioxide, producing sulfuric and nitric acids.
- **Greenhouse Effect & Global Warming:** The trapping of infrared heat radiation by atmospheric gases (CO₂, CH₄, N₂O, water vapour), which raises Earth's average surface temperature.
- **Ozone Depletion:** The destruction of stratospheric ozone shield molecules caused by pollutants like chlorofluorocarbons (CFCs).
- **Industrial Emission Control Devices:** Equipment such as scrubbers and electric precipitators installed in factory chimneys to reduce harmful industrial emissions.